

E-Commerce Sales Analysis — Full Jupyter Notebook Project

Project: E-Commerce Sales Analysis

Type: Data Analytics / Exploratory Data Analysis (EDA)

This notebook demonstrates a complete analytics workflow using Python, Pandas, NumPy and Matplotlib.

1. Problem Statement

An e-commerce company wants to understand its sales performance. The analysis covers:

- Revenue and profit
- Category performance
- Monthly trends
- Regional performance
- Top products
- Payment methods
- Discount and profitability
- Correlation analysis
- Business recommendations

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("ecommerce_sales_dataset.csv",
parse_dates=["Order_Date"])
df["Profit"] = df["Revenue"] - df["Cost"]
df["Profit_Margin_Pct"] = df["Profit"] / df["Revenue"] * 100

pd.set_option("display.max_columns", None)
pd.set_option("display.float_format", lambda x: f"{x:,.2f}")
df.head()
```

2. Data Understanding

```
print("Rows:", len(df))
print("Columns:", len(df.columns))
print("Missing values:", int(df.isnull().sum().sum()))
print("Duplicate rows:", int(df.duplicated().sum()))
print("\nData types:")
print(df.dtypes)
```

```
Rows: 180
Columns: 14
Missing values: 0
Duplicate rows: 0

Data types:
Order_ID                object
Order_Date              datetime64[ns]
Category                object
Product                 object
Region                  object
Units                   int64
Unit_Price               int64
Discount_Pct            int64
Revenue                  float64
Cost                     float64
Payment_Method           object
Profit                   float64
Profit_Margin_Pct        float64
Month                    object
```

3. Descriptive Statistics

```
df.describe().T
```

4. Key Performance Indicators

```
total_revenue = df["Revenue"].sum()
total_profit = df["Profit"].sum()
total_units = df["Units"].sum()
average_order_value = df["Revenue"].mean()
profit_margin = total_profit / total_revenue * 100

print(f"Total Revenue: ₹{total_revenue:,.2f}")
print(f"Total Profit: ₹{total_profit:,.2f}")
print(f"Units Sold: {total_units:,}")
print(f"Average Order Value: ₹{average_order_value:,.2f}")
print(f"Profit Margin: {profit_margin:.2f}%")

Total Revenue: ₹1,302,689.30
Total Profit: ₹437,574.04
Units Sold: 534
Average Order Value: ₹7,237.16
Profit Margin: 33.59%
```

5. Category Analysis

```
category_summary = df.groupby("Category").agg(
    Revenue=("Revenue", "sum"),
```

```

    Profit=("Profit","sum"),
    Units=("Units","sum")
).sort_values("Revenue", ascending=False)

category_summary["Profit_Margin_%"] = (
    category_summary["Profit"] / category_summary["Revenue"] * 100
)
category_summary.round(2)

plt.figure(figsize=(9,5))
plt.bar(category_summary.index, category_summary["Revenue"])
plt.title("Revenue by Category")
plt.xlabel("Category")
plt.ylabel("Revenue")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()

```

6. Monthly Revenue Trend

```

df["Month"] = df["Order_Date"].dt.to_period("M").astype(str)
monthly_revenue = df.groupby("Month")["Revenue"].sum()
monthly_revenue

plt.figure(figsize=(9,5))
plt.plot(monthly_revenue.index, monthly_revenue.values, marker="o")
plt.title("Monthly Revenue Trend")
plt.xlabel("Month")
plt.ylabel("Revenue")
plt.xticks(rotation=30)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

```

7. Regional Analysis

```

region_revenue = df.groupby("Region")
["Revenue"].sum().sort_values(ascending=False)
region_revenue

plt.figure(figsize=(8,5))
plt.bar(region_revenue.index, region_revenue.values)
plt.title("Revenue by Region")
plt.xlabel("Region")
plt.ylabel("Revenue")
plt.tight_layout()
plt.show()

```

8. Top 10 Products

```
top_products = df.groupby("Product").agg(  
    Revenue=("Revenue", "sum"),  
    Units=("Units", "sum")  
) .sort_values("Revenue", ascending=False).head(10)  
top_products  
  
plt.figure(figsize=(10,6))  
plt.barh(top_products.index[::-1], top_products["Revenue"][::-1])  
plt.title("Top 10 Products by Revenue")  
plt.xlabel("Revenue")  
plt.tight_layout()  
plt.show()
```

9. Payment Method Analysis

```
payment_revenue = df.groupby("Payment_Method")  
["Revenue"].sum().sort_values(ascending=False)  
payment_revenue
```

10. Discount and Profitability

```
discount_summary = df.groupby("Discount_Pct").agg(  
    Revenue=("Revenue", "sum"),  
    Profit=("Profit", "sum"),  
    Orders=("Order_ID", "count")  
)  
discount_summary["Profit_Margin_Pct"] = discount_summary["Profit"] /  
discount_summary["Revenue"] * 100  
discount_summary.round(2)  
  
plt.figure(figsize=(8,5))  
plt.scatter(df["Discount_Pct"], df["Profit"])  
plt.title("Discount Percentage vs Profit")  
plt.xlabel("Discount (%)")  
plt.ylabel("Profit")  
plt.grid(True, alpha=0.3)  
plt.tight_layout()  
plt.show()
```

11. Correlation Analysis

```
numeric_cols =  
["Units", "Unit_Price", "Discount_Pct", "Revenue", "Cost", "Profit", "Profit  
_Margin_Pct"]  
correlation = df[numeric_cols].corr()  
correlation.round(2)
```

```
plt.figure(figsize=(9,7))
plt.imshow(correlation, aspect="auto")
plt.colorbar(label="Correlation")
plt.xticks(range(len(correlation.columns)), correlation.columns,
rotation=45, ha="right")
plt.yticks(range(len(correlation.columns)), correlation.columns)
plt.title("Correlation Matrix")
plt.tight_layout()
plt.show()
```

12. Key Business Findings

- **Highest revenue category:** Electronics
- **Highest revenue region:** West
- **Highest revenue product:** Bluetooth Speaker
- **Highest revenue month:** 2026-02
- **Highest revenue payment method:** Cash on Delivery
- **Overall profit margin:** 33.59%

13. Recommendations

1. Focus inventory planning on high-revenue categories.
2. Strengthen marketing in strong-performing regions and investigate weaker regions.
3. Promote top products through bundles and cross-selling.
4. Monitor discount levels because discounts can affect profitability.
5. Support popular payment methods while retaining alternatives.
6. Use monthly trends to plan inventory and promotional campaigns.

Conclusion

This project demonstrates a complete EDA workflow: data loading, validation, KPI calculation, aggregation, visualization, correlation analysis, insights and business recommendations.

The synthetic dataset can be replaced with real company transaction data without changing the overall analysis structure.